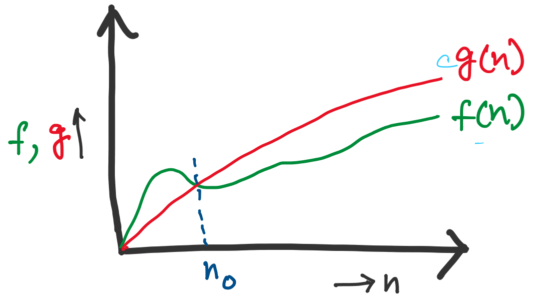
Growth of Functions

Wednesday, 29 July 2026 7:38 PM

A function $f(n)$ is said to be $O(g(n))$ if:

$$f(n) \le c\,g(n) \quad \forall n \geq n_0 \geq 0,\ \exists c > 0$$



$f(n) = O(g(n))$

– Implications?

1. Prove that: for any polynomial $f(n)$ of degree d, we have $f(n) = O(n^d)$

2. Prove that: $\underbrace{\frac{1}{2}n^2 - 3n}_{f(n)}$ is not $O(\underbrace{n}_{g(n)})$

* In the expression $f(n) = O(g(n))$, the '=' is not an "equality" sign.

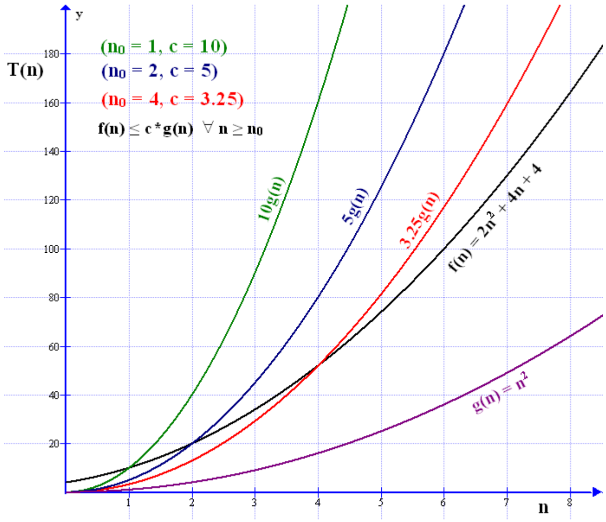
$2n^2 = O(n^2)$ ✗

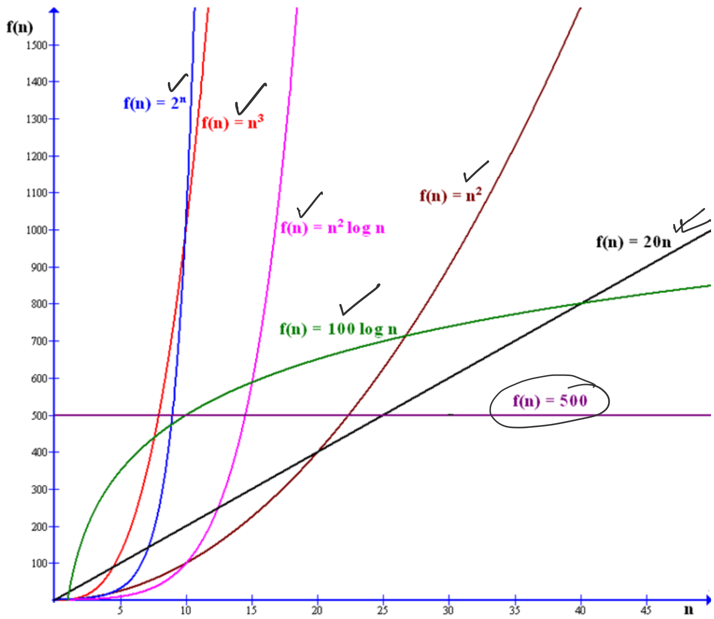
$3n^2 + 2 = O(n^2)$

$n^2 - 7n + 6 = O(n^2)$

$\in$

* Even if $f(n) = O(g(n))$, it is possible that $f(n) > g(n)$

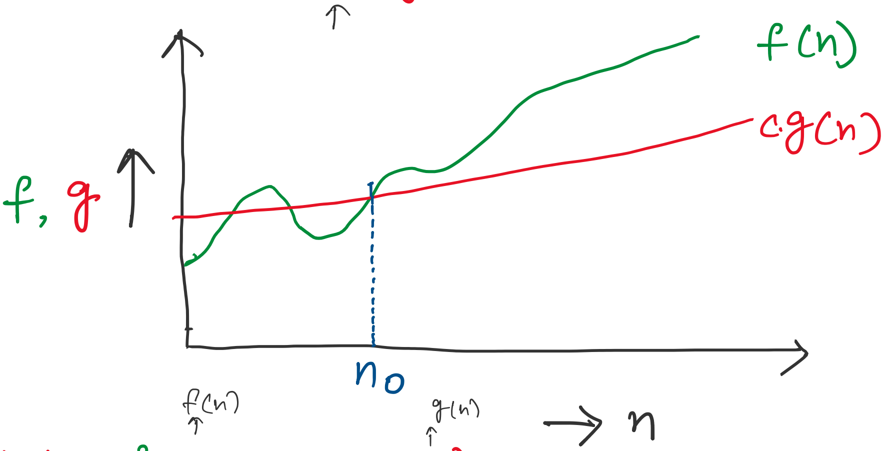




Ω- Notation

A function $f(n)$ is said to be $\Omega(g(n))$ if

$$f(n) \geq c\,g(n) \quad \forall n \geq n_0 > 0, \ \exists c > 0$$

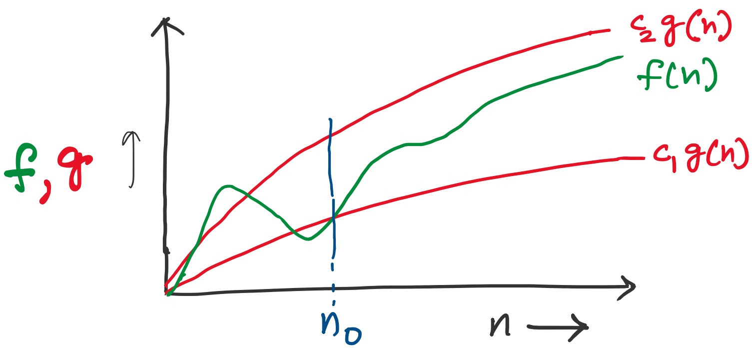


1. Prove that $\overset{f(n)}{3n^2+7n+8}$ is $\Omega(\overset{g(n)}{n^2})$

Θ - Notation

A function $f(n)$ is said to be $\Theta(g(n))$ if $f(n)$ is both $O(n)$ and $\Omega(n)$. Another way of saying this is:

$$\exists c_1, c_2, n_0 > 0 \text{ such that } \forall n > n_0, \quad c_1 g(n) \leq f(n) \leq c_2 g(n)$$



1. Show that: $\frac{1}{2}n^2 - 3n = \Theta(n^2)$